

# Reporte de Análisis Físico

## Validación MC y Asimetría UMD/SD

### Configuración de la Simulación

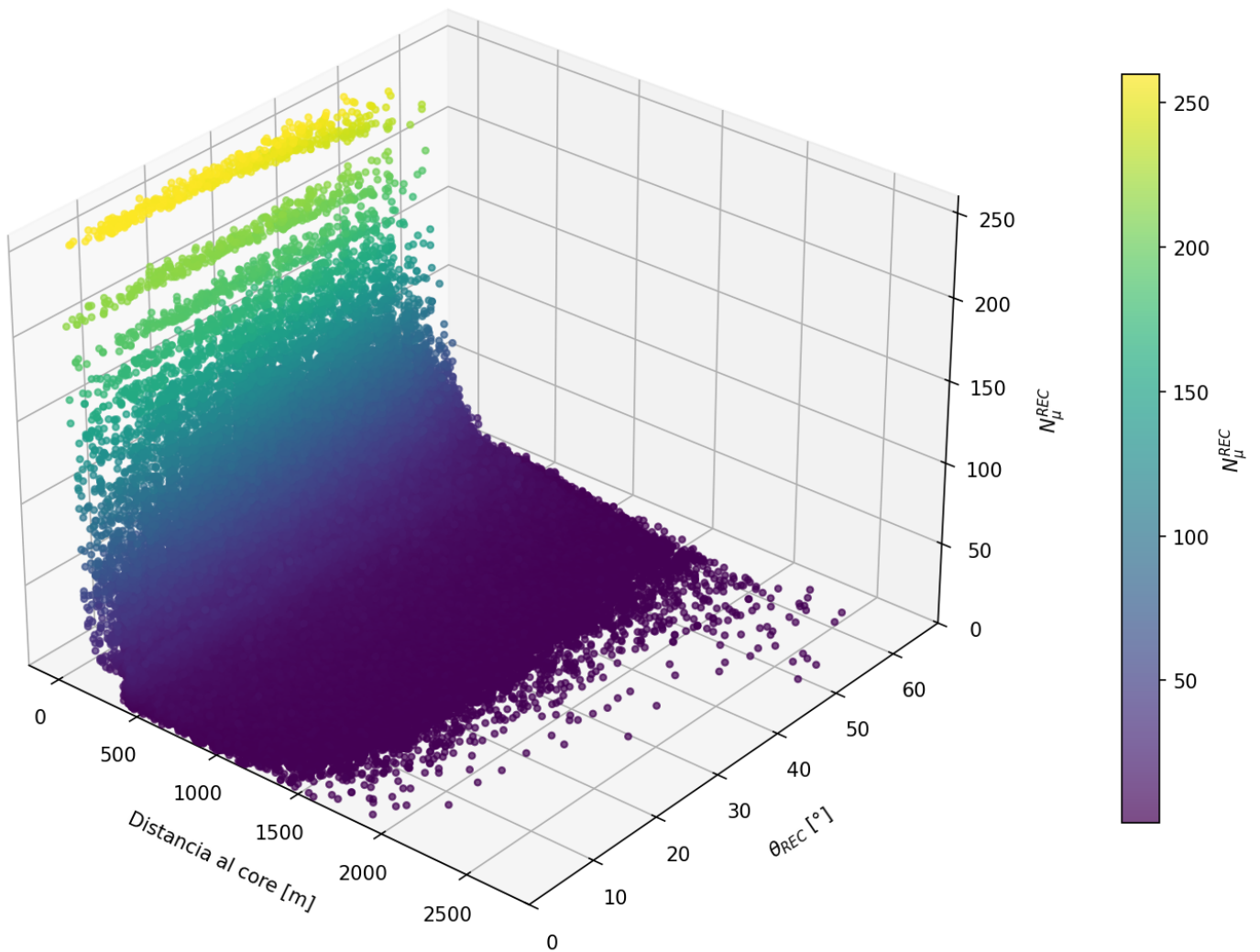
| Parámetro        | Valor               |
|------------------|---------------------|
| Modelo Hadrónico | SIBYLL 2.3d         |
| Primario         | Oxígeno             |
| Rango Energía    | 17.5 - 18.0 log(eV) |
| Carpeta          |                     |
| Eventos Totales  | 4,999               |
| Módulos Totales  | 1,649,367           |

## 1. Limpieza de Datos

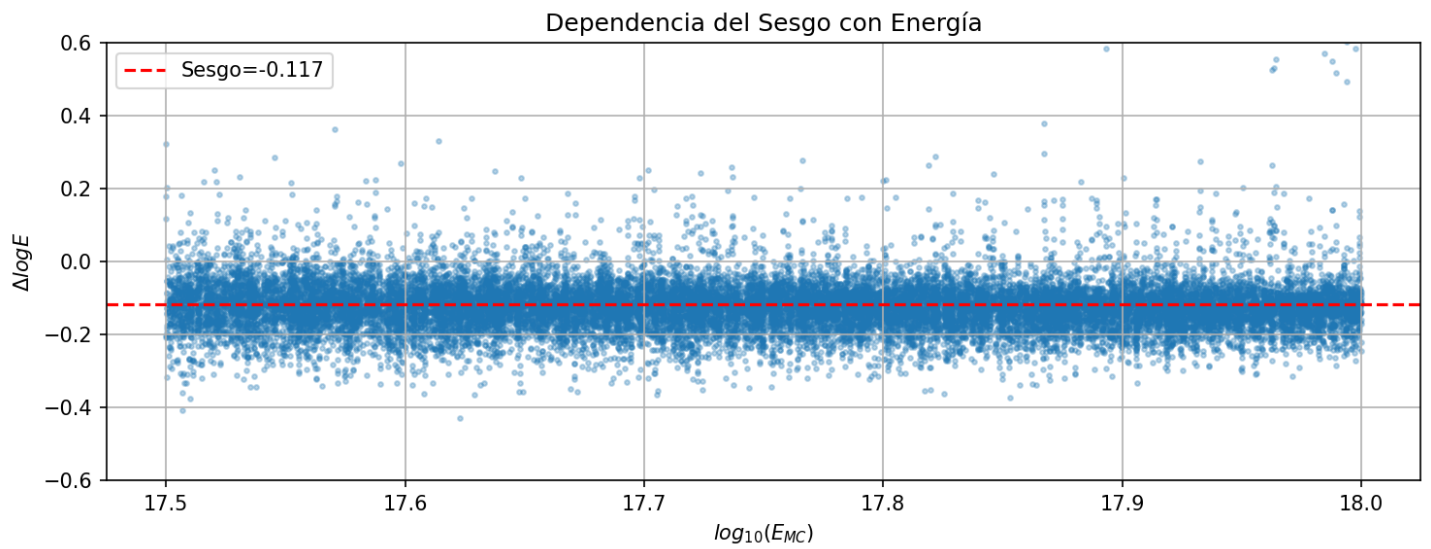
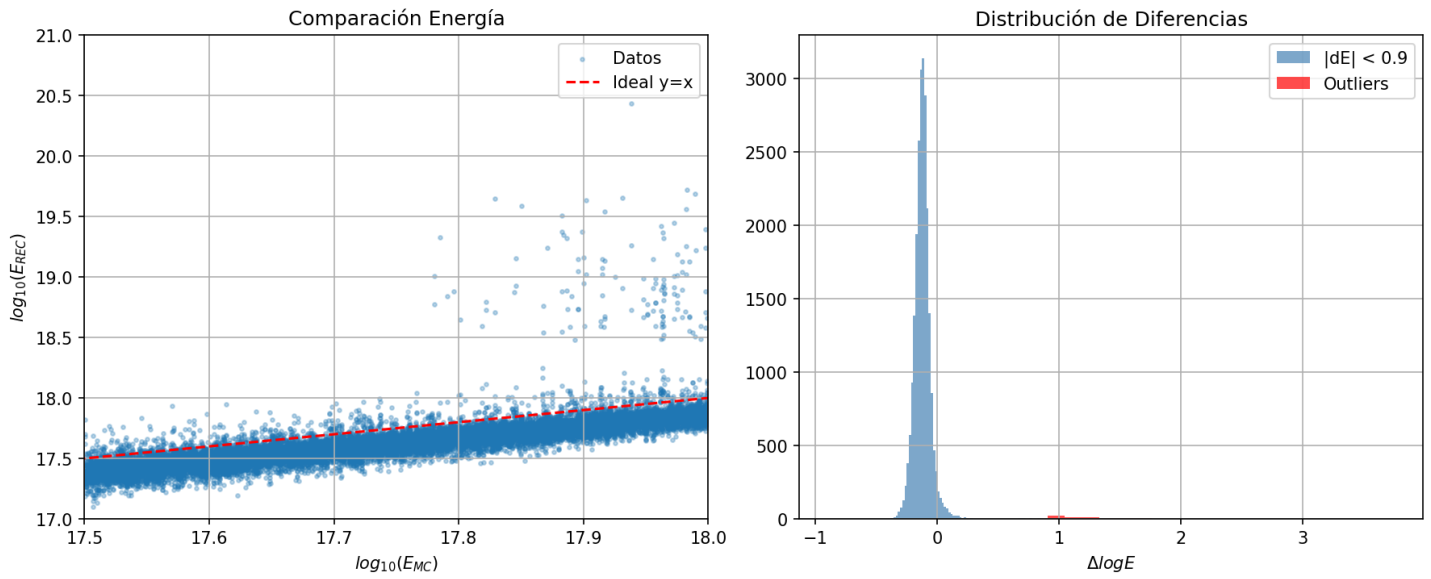
### REPORTE DE LIMPIEZA

| Métrica          | Valor                   |
|------------------|-------------------------|
| Filas Iniciales  | 1649367                 |
| Filas Eliminadas | 63492 (3.85%)           |
| Filas Finales    | 1585875                 |
| Causa Principal  | NaN en logE_REC (63492) |

## 2. Visualización General

$N_{\mu}^{REC}$  vs distancia y  $\theta$ 

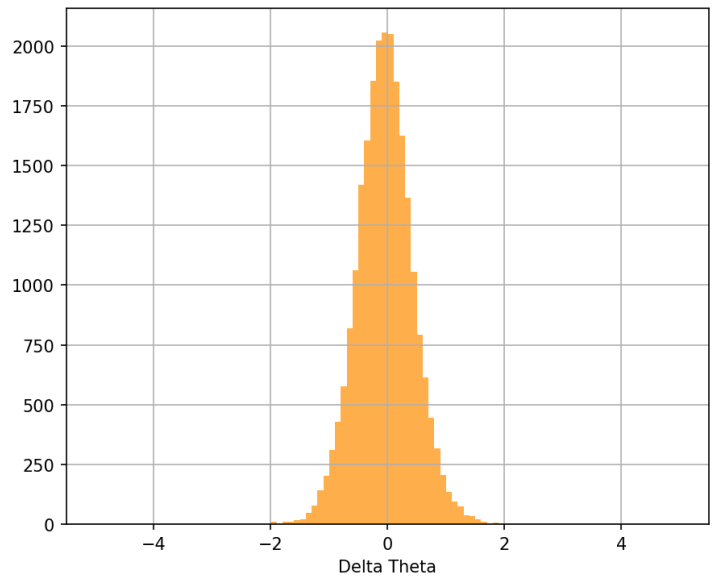
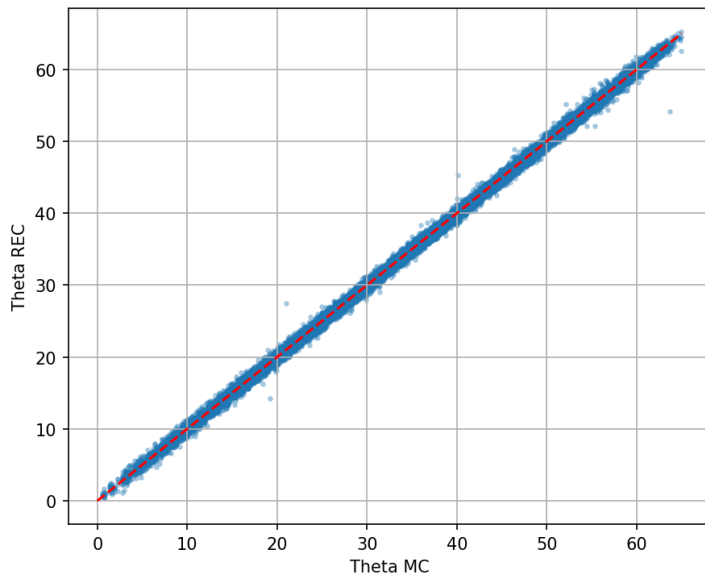
### 3. Resolución de Energía



Sesgo (Mean): -0.1170  
 Resolución (Std): 0.1109  
 Outliers (> 0.9): 70 (0.30%)

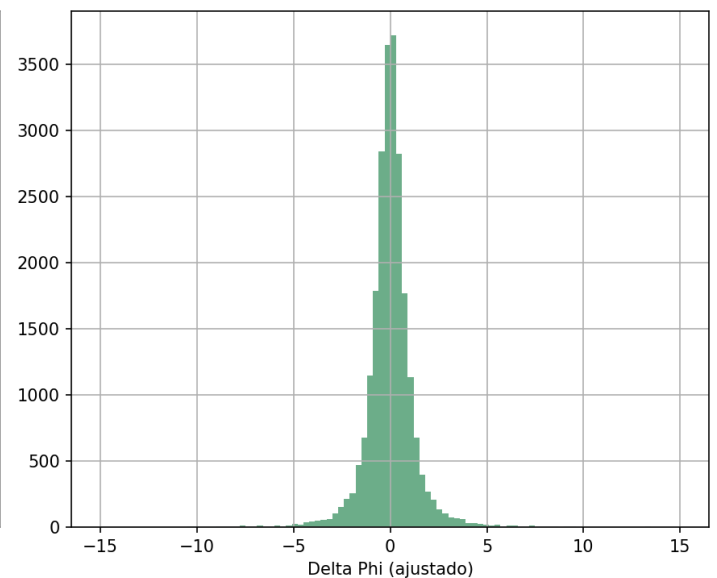
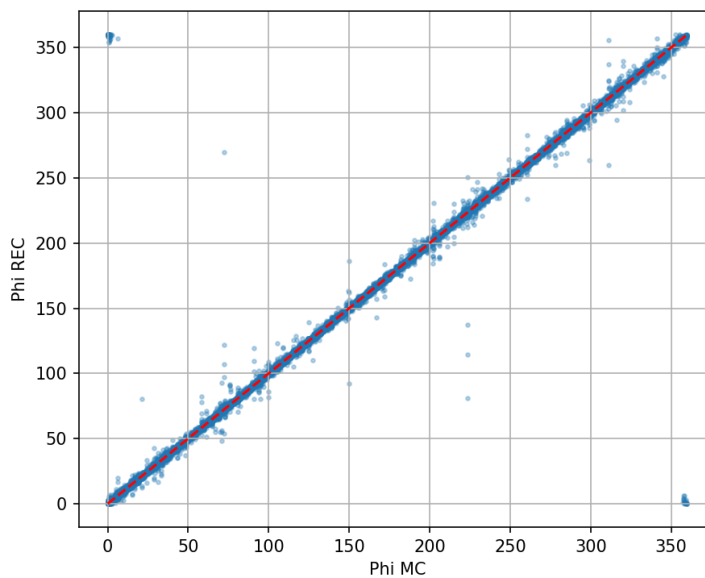
## 4. Resolución Angular

Theta (Cenital)



Sesgo:  $-0.0445$  deg  
Resolución:  $0.4889$  deg  
Outliers ( $> 10.0^\circ$ ): 0 (0.00%)

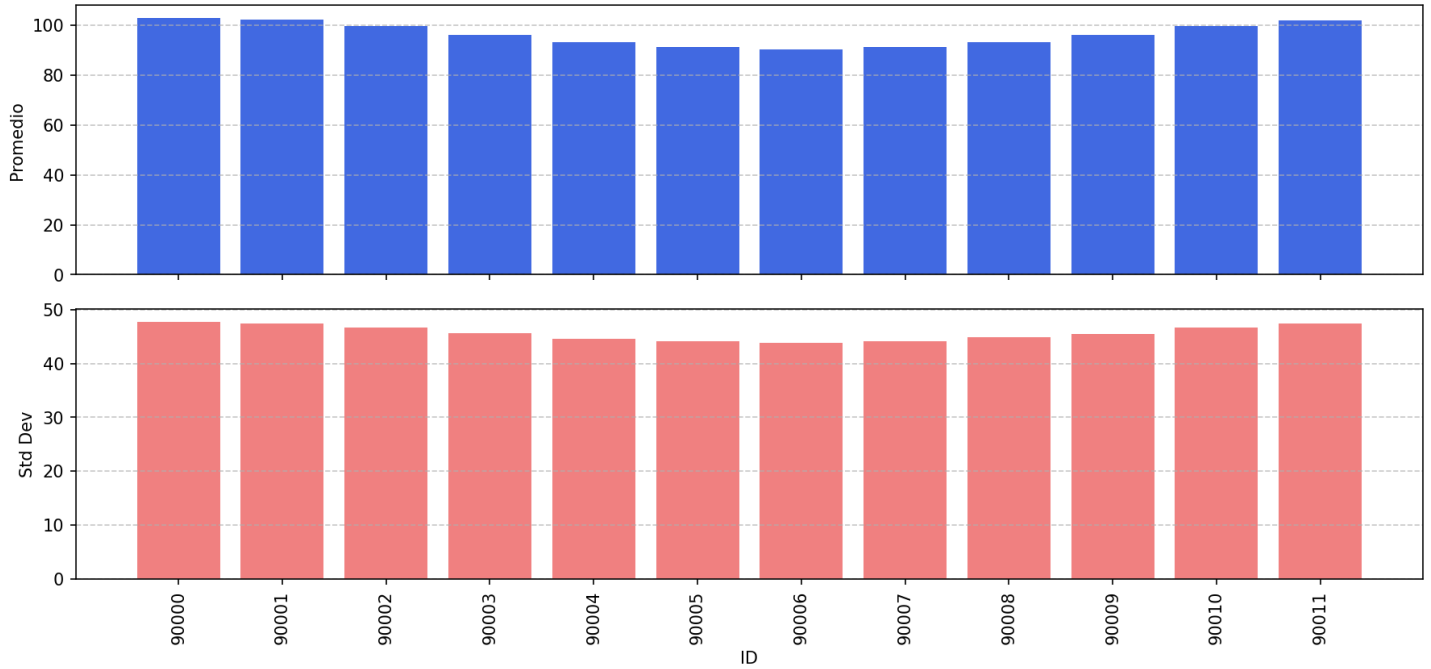
### Phi (Azimutal)



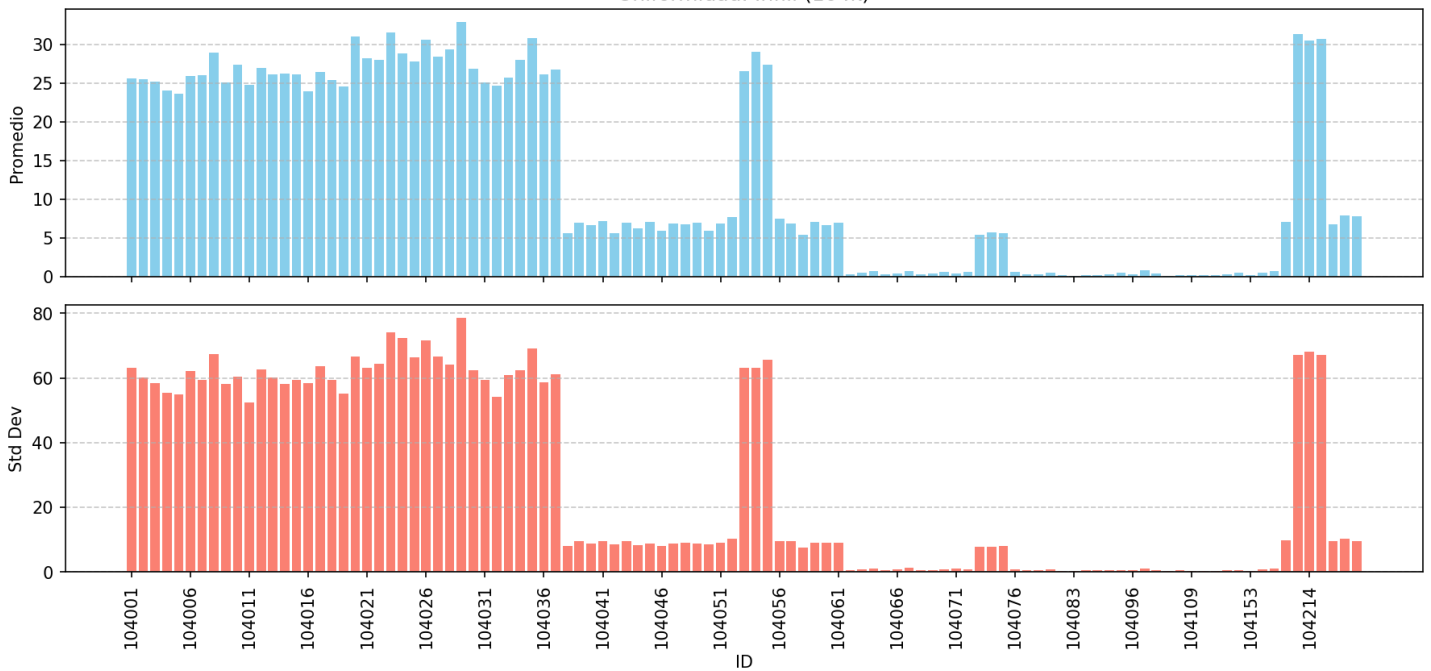
Sesgo (Median):  $-0.0013$  deg  
Resolución (MAD):  $0.5180$  deg  
Outliers ( $> 15.0^\circ$ ): 50 (0.21%)

## 5. Uniformidad por Counter

Uniformidad: Anillo Denso (90k)

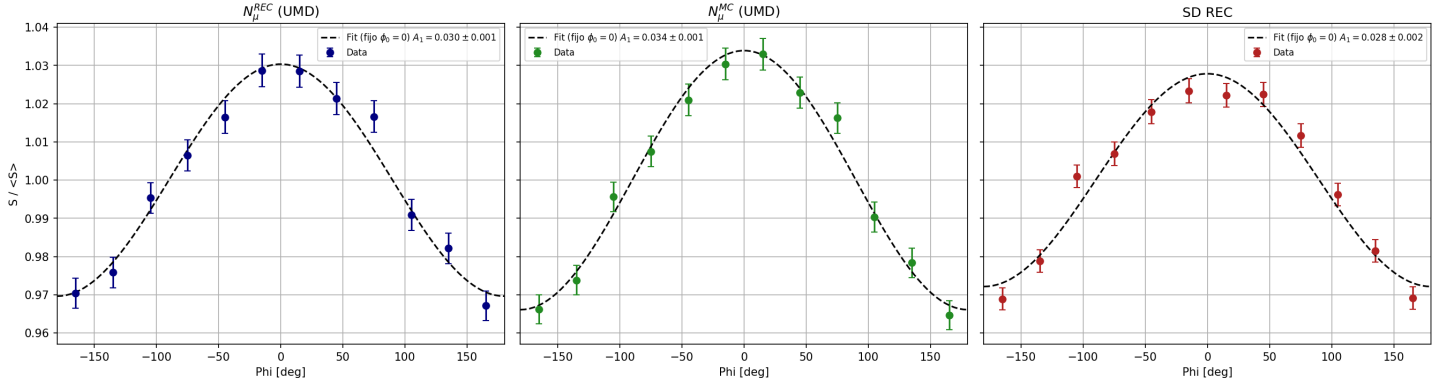


Uniformidad: Infill (104k)

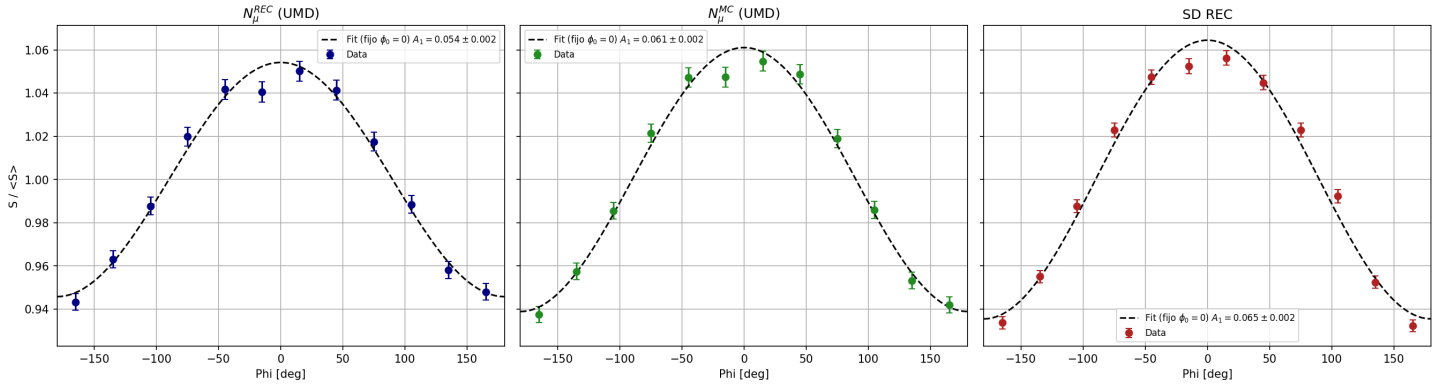


## 6. Asimetría Azimutal

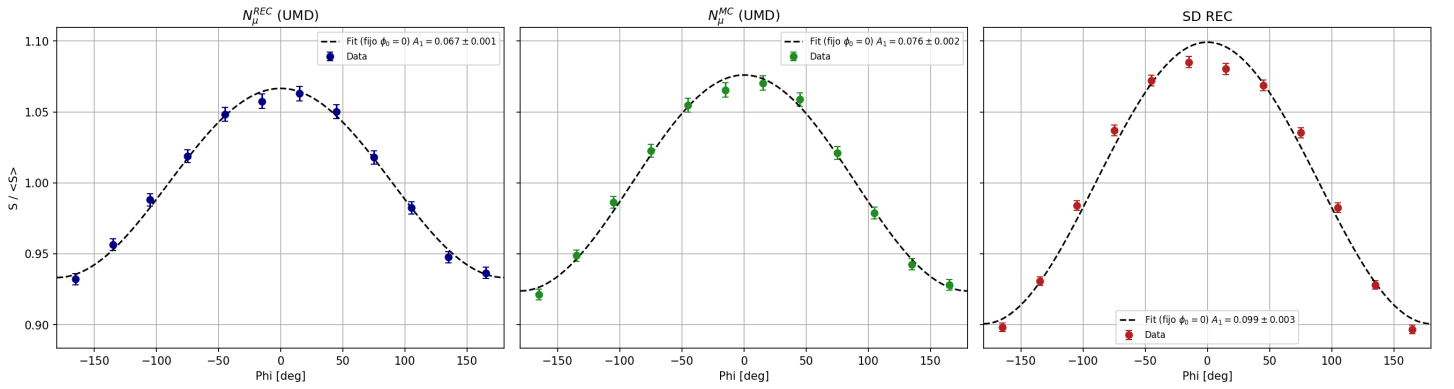
Bin 0: Theta 0.4 - 21.7 deg



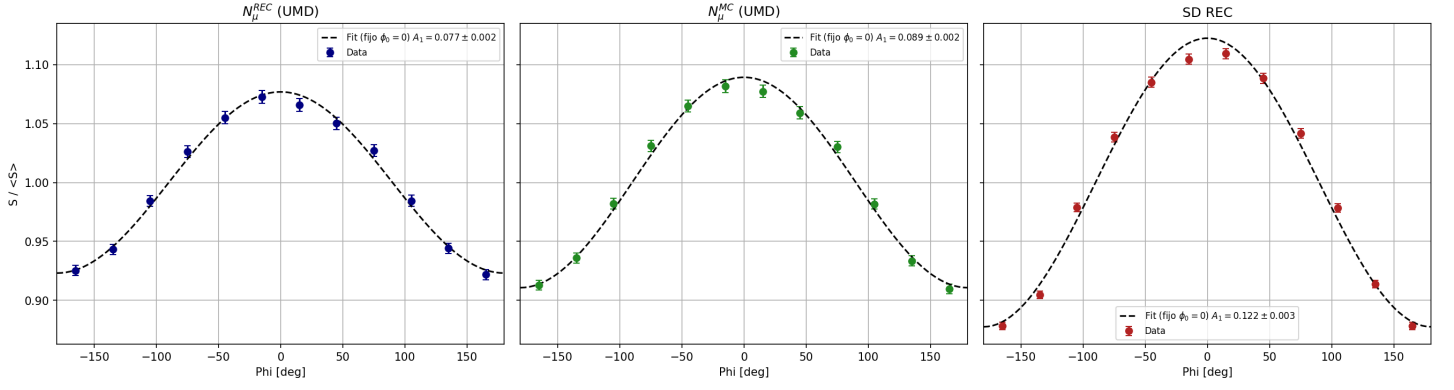
### Bin 1: Theta 21.7 - 31.6 deg



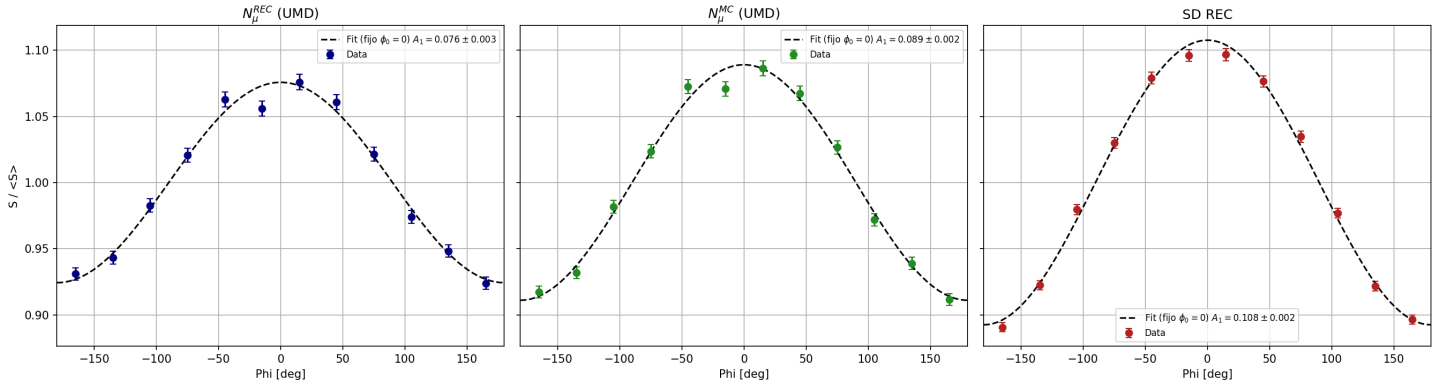
### Bin 2: Theta 31.6 - 39.9 deg



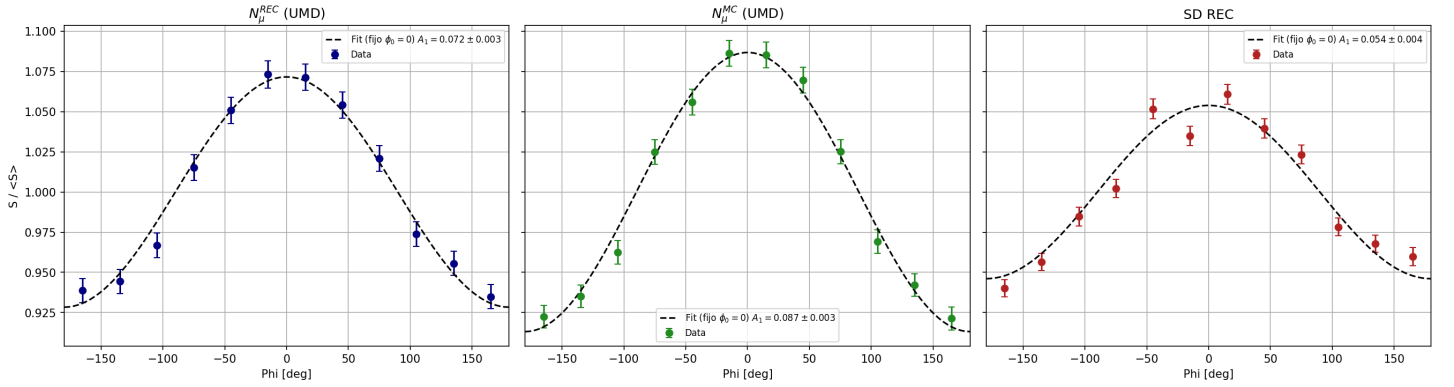
### Bin 3: Theta 39.9 - 47.7 deg



#### Bin 4: Theta 47.7 - 55.8 deg



#### Bin 5: Theta 55.8 - 65.0 deg



## 7. Resumen de Ajustes (Comparación)

Valores de Amplitud A1

| theta_range | A1_UMD_REC | A1_UMD_MC | A1_SD_REC |
|-------------|------------|-----------|-----------|
| 0.4-21.7    | 0.0304     | 0.0339    | 0.0278    |
| 21.7-31.6   | 0.0542     | 0.0611    | 0.0645    |
| 31.6-39.9   | 0.0667     | 0.0761    | 0.0992    |
| 39.9-47.7   | 0.0769     | 0.0893    | 0.1225    |
| 47.7-55.8   | 0.0757     | 0.0890    | 0.1076    |
| 55.8-65.0   | 0.0716     | 0.0869    | 0.0539    |

